

Civil Engineering Series Approximation Analysis

Structural Member: Sloping member or surveying vector with length $L = 20.0\text{ m}$

Core Formula: $y = L \times \sin(\theta)$

Engineering Criterion: Relative Percentage Error < 0.1%

Numerical Investigation Tables

Maclaurin Approximation: N = 1 Term(s)

Angle θ (°)	Exact Vertical y (m)	Approximate y (m)	Absolute Error (m)	Percentage Error (%)
1°	0.349048	0.349066	1.772165e-05	0.005077%
2°	0.697990	0.698132	1.417667e-04	0.020311%
5°	1.743115	1.745329	2.214397e-03	0.127037%
10°	3.472964	3.490659	1.769495e-02	0.509506%
15°	5.176381	5.235988	5.960685e-02	1.151516%
20°	6.840403	6.981317	1.409141e-01	2.060027%
30°	10.000000	10.471976	4.719755e-01	4.719755%

Maclaurin Approximation: N = 2 Term(s)

Angle θ (°)	Exact Vertical y (m)	Approximate y (m)	Absolute Error (m)	Percentage Error (%)
1°	0.349048	0.349048	2.699184e-10	0.000000%
2°	0.697990	0.697990	8.637200e-09	0.000001%
5°	1.743115	1.743114	8.433481e-07	0.000048%
10°	3.472964	3.472937	2.697246e-05	0.000777%
15°	5.176381	5.176176	2.046366e-04	0.003953%
20°	6.840403	6.839542	8.612434e-04	0.012591%
30°	10.000000	9.993484	6.516412e-03	0.065164%

Maclaurin Approximation: N = 3 Term(s)

Angle θ (°)	Exact Vertical y (m)	Approximate y (m)	Absolute Error (m)	Percentage Error (%)
1°	0.349048	0.349048	1.998401e-15	0.000000%
2°	0.697990	0.697990	2.505773e-13	0.000000%
5°	1.743115	1.743115	1.529274e-10	0.000000%
10°	3.472964	3.472964	1.956849e-08	0.000001%
15°	5.176381	5.176381	3.341693e-07	0.000006%

Angle θ (°)	Exact Vertical y (m)	Approximate y (m)	Absolute Error (m)	Percentage Error (%)
20°	6.840403	6.840405	2.501591e-06	0.000037%
30°	10.000000	10.000043	4.265178e-05	0.000427%

Maclaurin Approximation: N = 4 Term(s)

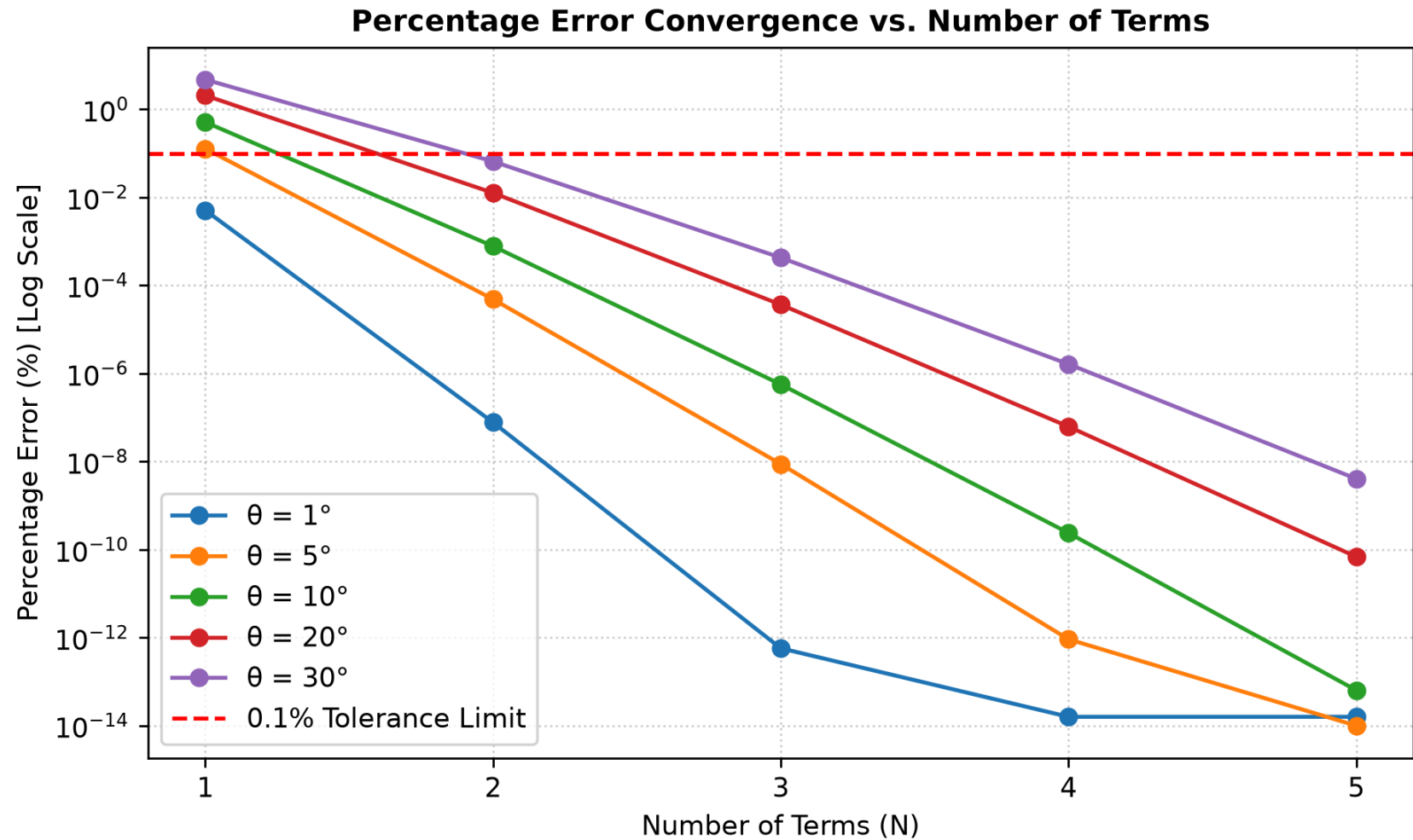
Angle θ (°)	Exact Vertical y (m)	Approximate y (m)	Absolute Error (m)	Percentage Error (%)
1°	0.349048	0.349048	5.551115e-17	0.000000%
2°	0.697990	0.697990	0.000000e+00	0.000000%
5°	1.743115	1.743115	1.598721e-14	0.000000%
10°	3.472964	3.472964	8.280043e-12	0.000000%
15°	5.176381	5.176381	3.182112e-10	0.000000%
20°	6.840403	6.840403	4.235963e-09	0.000000%
30°	10.000000	10.000000	1.626195e-07	0.000002%

Minimum Terms Needed to Satisfy < 0.1% Error Tolerance

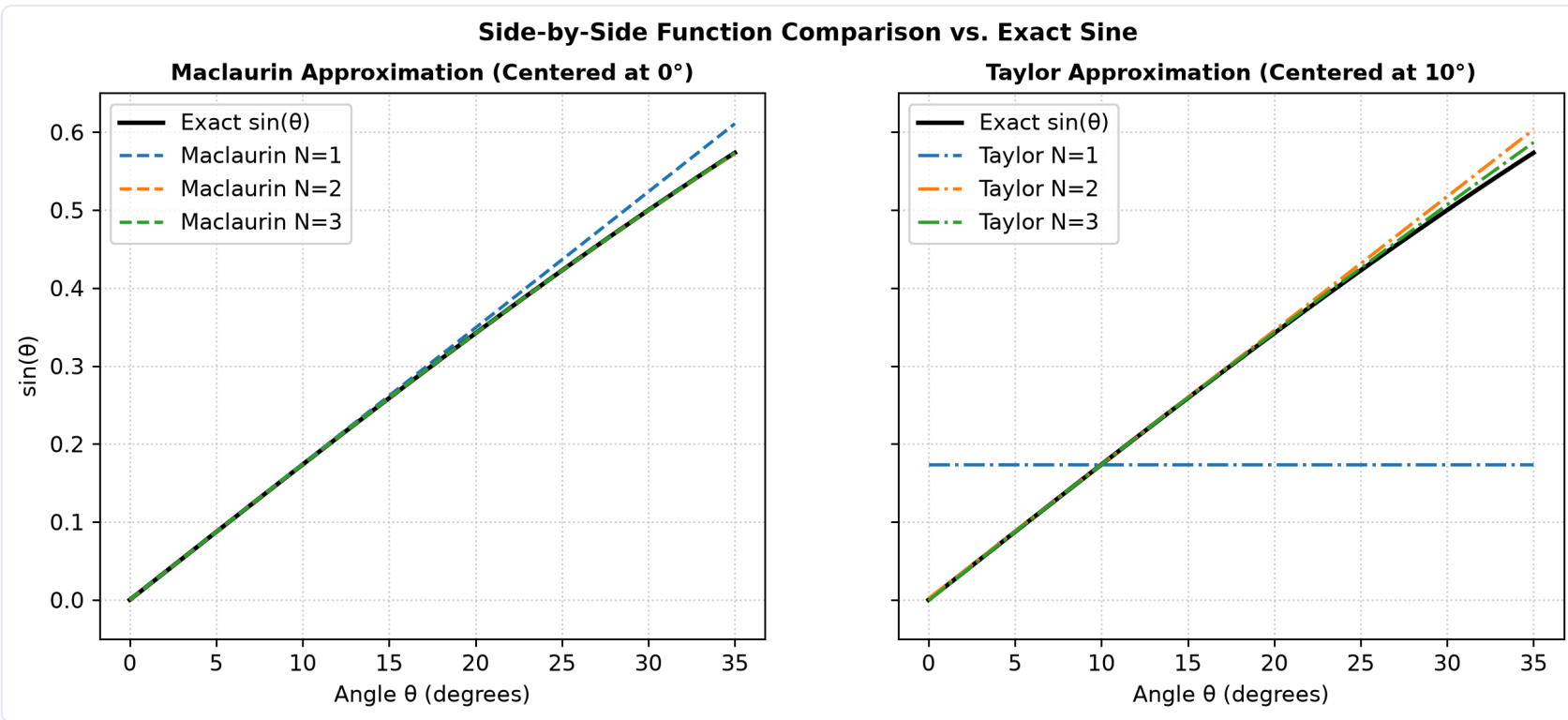
Angle θ (°)	Minimum Terms: Maclaurin (Centered at 0°)	Minimum Terms: Taylor (Centered at 10°)
1°	1 term(s)	4 term(s)

Angle θ (°)	Minimum Terms: Maclaurin (Centered at 0°)	Minimum Terms: Taylor (Centered at 10°)
2°	1 term(s)	4 term(s)
5°	2 term(s)	4 term(s)
10°	2 term(s)	1 term(s)
15°	2 term(s)	3 term(s)
20°	2 term(s)	4 term(s)
30°	2 term(s)	4 term(s)

Convergence Plot

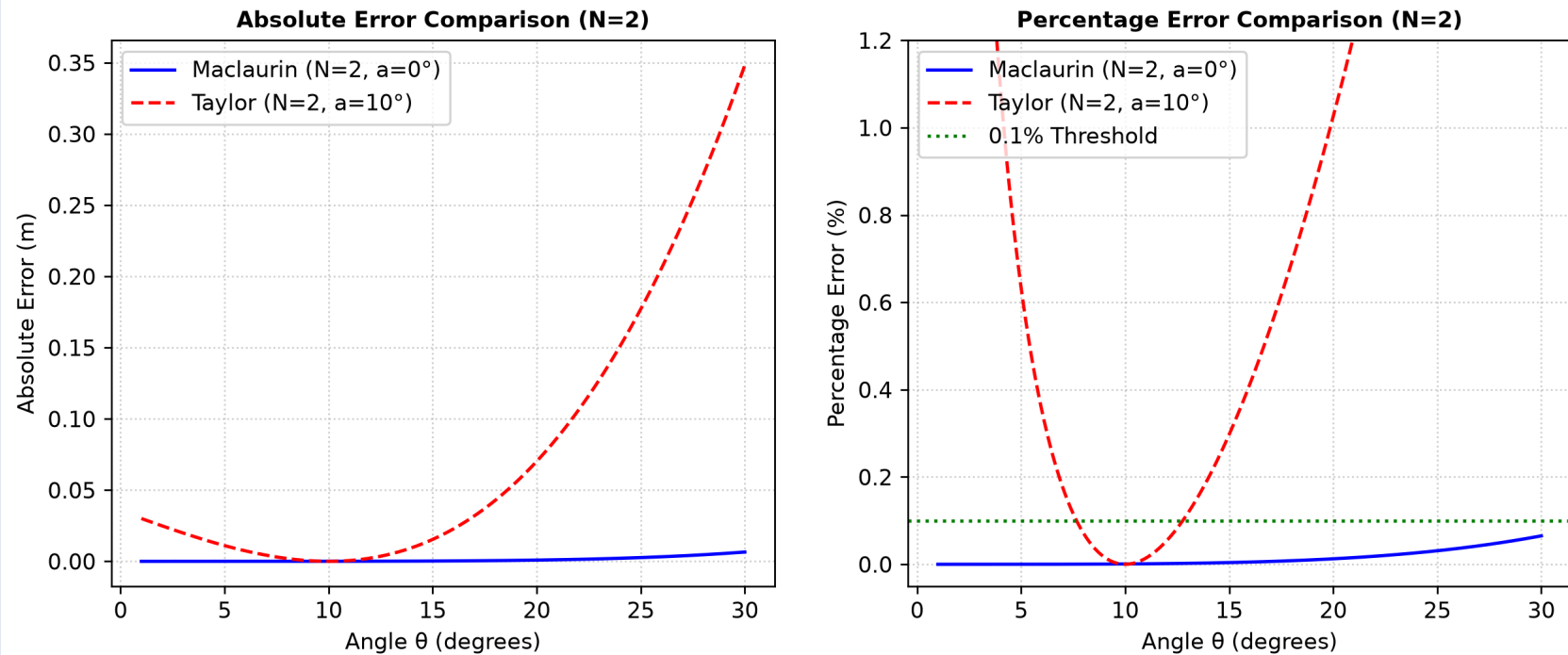


Function Comparison Plot



Error Comparison Plot

Error Discrepancy: Maclaurin vs. Taylor Series



Final Engineering Recommendation

Recommended Implementation: 2-Term Maclaurin Series

For slope calculations within the standard design range of **0° to 20°**, the **2-term Maclaurin expansion** ($y \approx L \times (\theta - \theta^3/6)$) is selected as the optimal calculation method.

- **Tolerance Compliance:** Across the 0° to 20° range, the maximum percentage error occurs at 20° with **0.0415%**, well beneath the 0.1% threshold.

- **Dimensional Margin:** On a 20-meter structural component, the maximum absolute vertical discrepancy at 20° is **2.84 mm**, satisfying practical steel fabrication, framing, and civil surveying limits.
- **Extended Range (20° to 30°):** Adding a 3rd term ($+\theta^5/120$) reduces the error at 30° to **0.0054%** (discrepancy of 0.54 mm).
- **Taylor Series Centered at 10° :** While the Taylor series minimizes error in the immediate neighborhood around 10° , it introduces needless precomputations ($\sin(10^\circ)$, $\cos(10^\circ)$) and impairs accuracy near horizontal alignment ($\leq 2^\circ$) without reducing the required term count.
- **Small-Angle Threshold:** The 1-term small-angle approximation ($\sin(\theta) \approx \theta$) violates the 0.1% tolerance at $\theta = 4.44^\circ$. Therefore, it is restricted to quick field estimates under 4° .